

Sweet potato: a review of its past, present, and future role in human nutrition.



The overall objective of this chapter is to review the past, present, and future role of the sweet potato (*Ipomoea batatas* [L.] Lam) in human nutrition. Specifically, the chapter describes the role of the sweet potato in human diets; outlines the biochemical and nutritional composition of the sweet potato with emphasis on its beta-carotene and anthocyanin contents; highlights sweet potato utilization, and its potential as value-added products in human food systems; and demonstrates the potential of the sweet potato in the African context. Early records have indicated that the sweet potato is a staple food source for many indigenous populations in Central and South Americas, Ryukyu Island, Africa, the Caribbean, the Maori people, Hawaiians, and Papua New Guineans. Protein contents of sweet potato leaves and roots range from 4.0% to 27.0% and 1.0% to 9.0%, respectively. The sweet potato could be considered as an excellent novel source of natural health-promoting compounds, such as beta-carotene and anthocyanins, for the functional food market. Also, the high concentration of anthocyanin and beta-carotene in sweet potato, combined with the high stability of the color extract make it a promising and healthier alternative to synthetic coloring agents in food systems. Starch and flour processing from sweet potato can create new economic and employment activities for farmers and rural households, and can add nutritional value to food systems. Repositioning sweet potato production and its potential for value-added products will contribute substantially to utilizing its benefits and many uses in human food systems. Multidisciplinary, integrated research and development activities aimed at improving production, storage, postharvest and processing technologies, and quality of the sweet potato and its potential value-added products are critical issues, which should be addressed globally.